



INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA
SALT LAKE

Affiliated to UEM Kolkata and Approved by AICTE



TERM EXAMINATION

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Stream: CSBS

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Sec.:

Class Roll No.: 11

Paper Name: COMPUTER ORGANISATION & ARCHITECTURE Paper Code: PCCCS40

Date: 09.02.2026

Instruction to Candidates

1. All answers must be written ON BOTH SIDES of the paper.
2. No loose paper will be provided for scribbling for rough work and no paper should be brought for this purpose. No page should be torn. The back cover may be used for rough work.
3. Candidates are required to write answer in their own words.
4. This answer book must be submitted to the invigilator-in-charge, even if blank.
5. A candidate cannot leave the Examination Hall within 45 minutes of the commencement of examination.
6. Unfair means in any form if adopted by a candidate would be liable him/her to be expelled immediately from the Examination Hall.

For Examiner's use Only

Q. No.	Marks
Total	26

Tuhina Karan,

Student's Signature

[Signature]

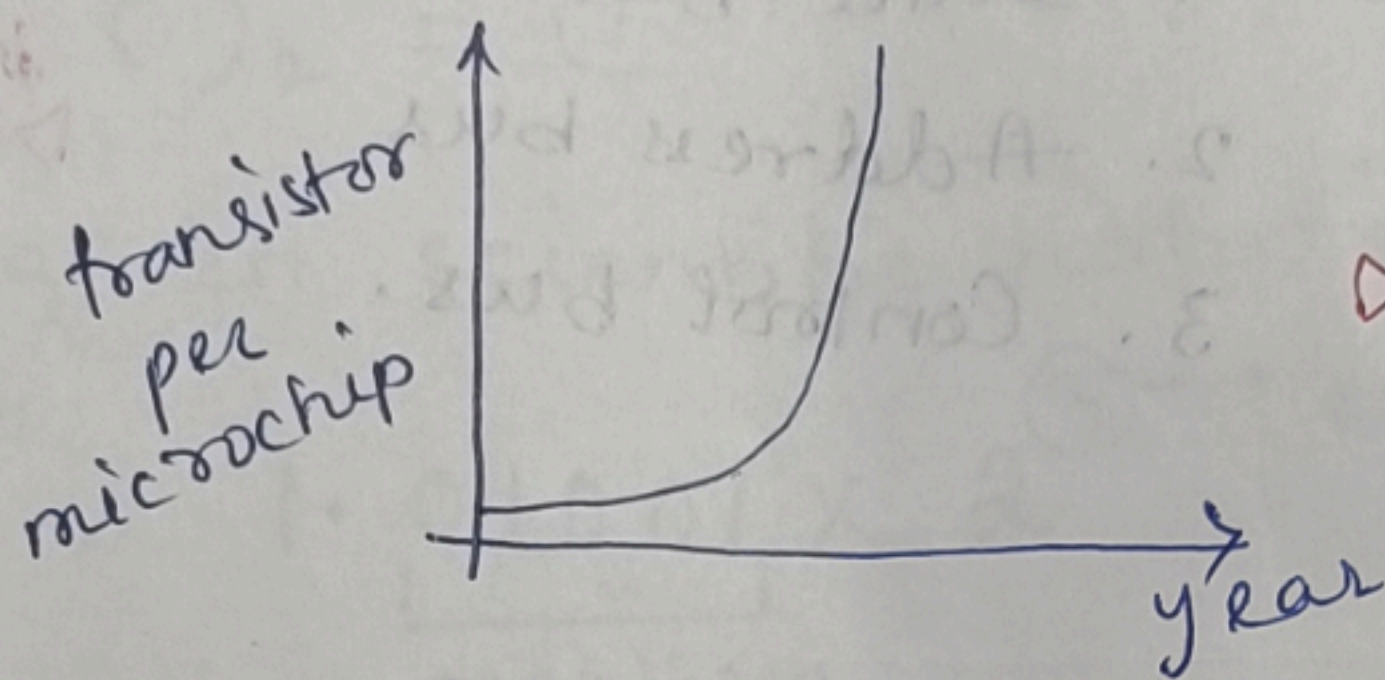
Invigilator's Signature

[Signature]

Examiner's Signature

SECTION - A

1) Moore's law states that the number of transistors per chip grows every 18 to 24 months, leading to exponential growth in the computing power and less cost.



2) Opcode is the set of instructions ~~written~~ in binary is stored in the memory ~~block~~ of the computer.

4) a) $(70)_{10} \rightarrow (?)_{16}$ $\begin{array}{r} 16 \overline{) 70} \\ \underline{64} \\ 6 \end{array}$

$\begin{array}{r} 2 \overline{) 70} \\ \underline{64} \\ 6 \end{array}$

$\begin{array}{r} 2 \overline{) 35} \\ \underline{32} \\ 3 \end{array}$

$\begin{array}{r} 2 \overline{) 17} \\ \underline{16} \\ 1 \end{array}$

$\begin{array}{r} 2 \overline{) 8} \\ \underline{8} \\ 0 \end{array}$

$\begin{array}{r} 2 \overline{) 4} \\ \underline{4} \\ 0 \end{array}$

$(46)_{16}$

b) $(130)_{10} \rightarrow (?)_{16}$

$\begin{array}{r} 16 \overline{) 130} \\ \underline{128} \\ 2 \end{array}$

$(82)_{16}$

c) $(1348)_{10} \rightarrow (?)_{16}$

$\begin{array}{r} 16 \overline{) 1348} \\ \underline{1280} \\ 68 \end{array}$

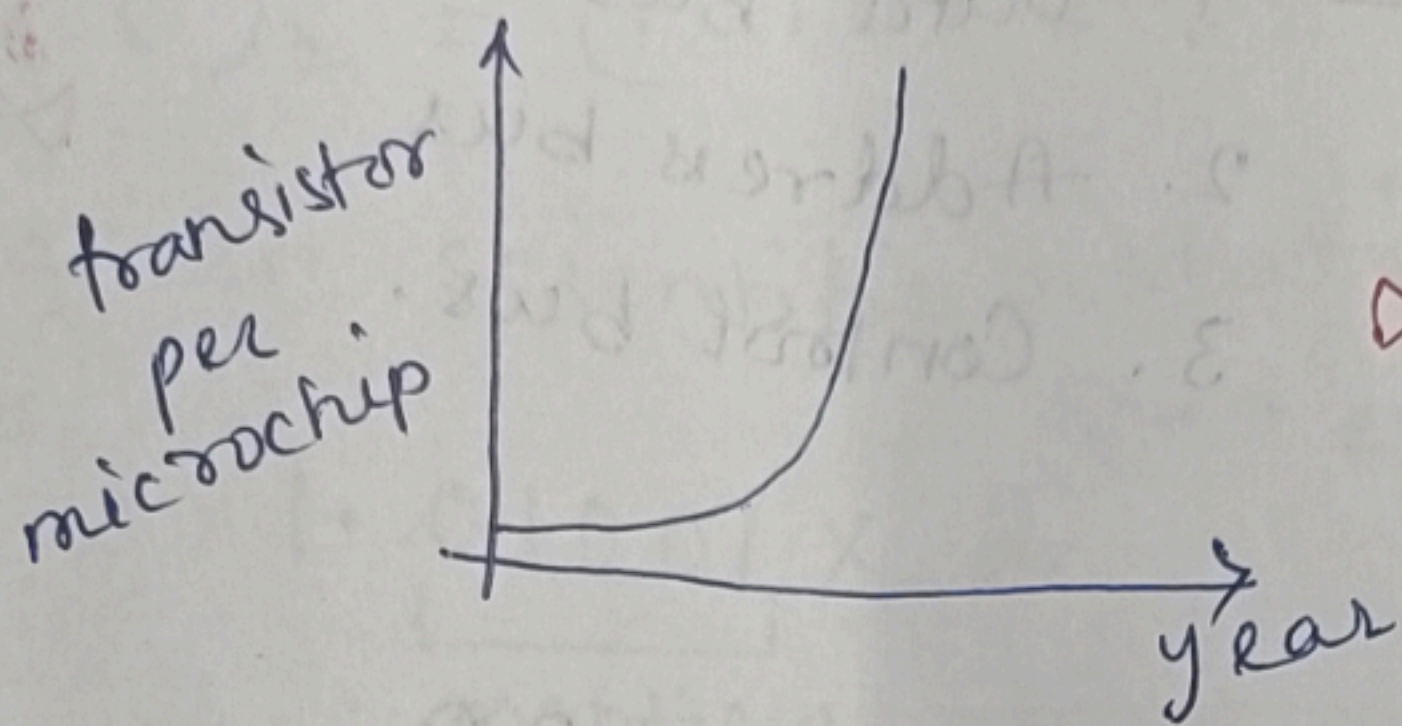
$\begin{array}{r} 16 \overline{) 68} \\ \underline{64} \\ 4 \end{array}$

$\begin{array}{r} 16 \overline{) 4} \\ \underline{4} \\ 0 \end{array}$

$(844)_{16}$

SECTION - A

1) Moore's law states that the number of transistors per chip grows every 18 to 24 months, leading to exponential growth in the computing power and less cost.



2) Opcode is the set of instructions ~~written~~ in binary and is stored in the memory ~~block~~ of the computer.

4) a) $(70)_{10} \rightarrow (?)_{16}$ $16 \overline{) 70} - 6$ $\uparrow$

$2 \overline{) 70} - 0$

$2 \overline{) 35} - 1$

$2 \overline{) 17} - 1$

$2 \overline{) 8} - 0$

$2 \overline{) 4}$

$(46)_{16}$

b) $(130)_{10} \rightarrow (?)_{16}$

$16 \overline{) 130} - 2$
 $16 \overline{) 8} - 8$

$= (82)_{16}$

c) $(1348)_{10} \rightarrow (?)_{16}$

$16 \overline{) 1348} - 4$

$16 \overline{) 84} - 4$

$16 \overline{) 5} - 5$

$(544)_{16}$

5) Bus is a component of the computer. It is used for transmission of instructions, data, addresses and control signals between the main memory, the processor and input-output devices.

Types of buses —

1. Data bus
2. Address bus
3. Control bus.

6) 1011011

1's complement = 0100100

2's complement = 1's complement + 1
= 0100101

SECTION-B

7) IEEE 754 format floating point numbers in single precision :- (32 bits)

sign bit	exponent	Mantissa
----------	----------	----------

1 bit

8 bit

remaining

i.e. $32 - (1 + 8)$

bias exponent
+ power

23 bits.

$$= (2^{8-1} - 1) + \text{power}$$

$$= 127 + \text{power}$$

IEEE 754 format for floating point numbers in double precision :- (64 bits)

sign bit	exponent	Mantissa
----------	----------	----------

1 bit

11 bits

remaining i.e.

$$64 - (1 + 11)$$

$$= 52 \text{ bits}$$

11 bits bias exponent + power

Convert in binary.

$$2 \overline{) 10 - 0}$$

$$2 \overline{) 5 - 1}$$

$$2 \overline{) 2 - 0}$$

1

$$= (1010)_2$$

$$25 \times 2 = 0.5$$

$$.5 \times 2 = 1.0$$

↓

$$(10.25)_2 = (1010.01)_2$$

Expressing in scientific notation, we get,

$$(-1)^0 \times 1.01001 \times 2^3 \rightarrow \text{exponent.}$$

↑
mantissa

+ve sign

single precision

IEEE754 representation format :-

0

↑

sign bit

(0 for +ve sign)

10000010

↑
exponent

$$127 + 3$$

$$= 130$$

010010000000000000000000

mantissa

(remaining 23 bits with 0 padding)

128 64 32 16 8 4 2 1
1 0 0 0 0 0 1

5

e) Non-restoring division :-

$$\begin{array}{r} 5 \div 3 \\ \hline Q \quad M \end{array}$$

$$\begin{array}{l} Q(\text{dividend}) = 0101 \\ M(\text{divisor}) = 0011 \end{array} \left. \vphantom{\begin{array}{l} Q(\text{dividend}) \\ M(\text{divisor}) \end{array}} \right\} \begin{array}{l} 4 \text{ bits, hence} \\ 4 \text{ stages.} \end{array}$$

Operation	Accumulator (A)	dividend (Q)	divisor (m)
Initialize	0 0 0 0	0 1 0 1	m = 0 0 1 1 -m = 1 1 0 1

Stage 1:

Left shift

sub m

$$\begin{array}{r} \boxed{0} \ 0 \ 0 \ 0 \quad 1 \ 0 \ 1 \text{ ---} \\ 1 \ 1 \ 0 \ 1 \\ \hline \boxed{1} \ 1 \ 0 \ 1 \quad 1 \ 0 \ 1 \ 0 \end{array}$$

Stage 2:

Left shift

add m

$$\begin{array}{r} \boxed{1} \ 0 \ 1 \ 1 \quad 0 \ 1 \ 0 \text{ ---} \\ 0 \ 0 \ 1 \ 1 \\ \hline \boxed{1} \ 1 \ 1 \ 0 \quad 0 \ 1 \ 0 \ 0 \end{array}$$

Stage 3:

Left shift

add m

$$\begin{array}{r} \boxed{1} \ 1 \ 0 \ 0 \quad 1 \ 0 \ 0 \text{ ---} \\ 0 \ 0 \ 1 \ 1 \\ \hline \boxed{1} \ 1 \ 1 \ 1 \quad 1 \ 0 \ 0 \ 0 \end{array}$$

Stage 4:

Left shift

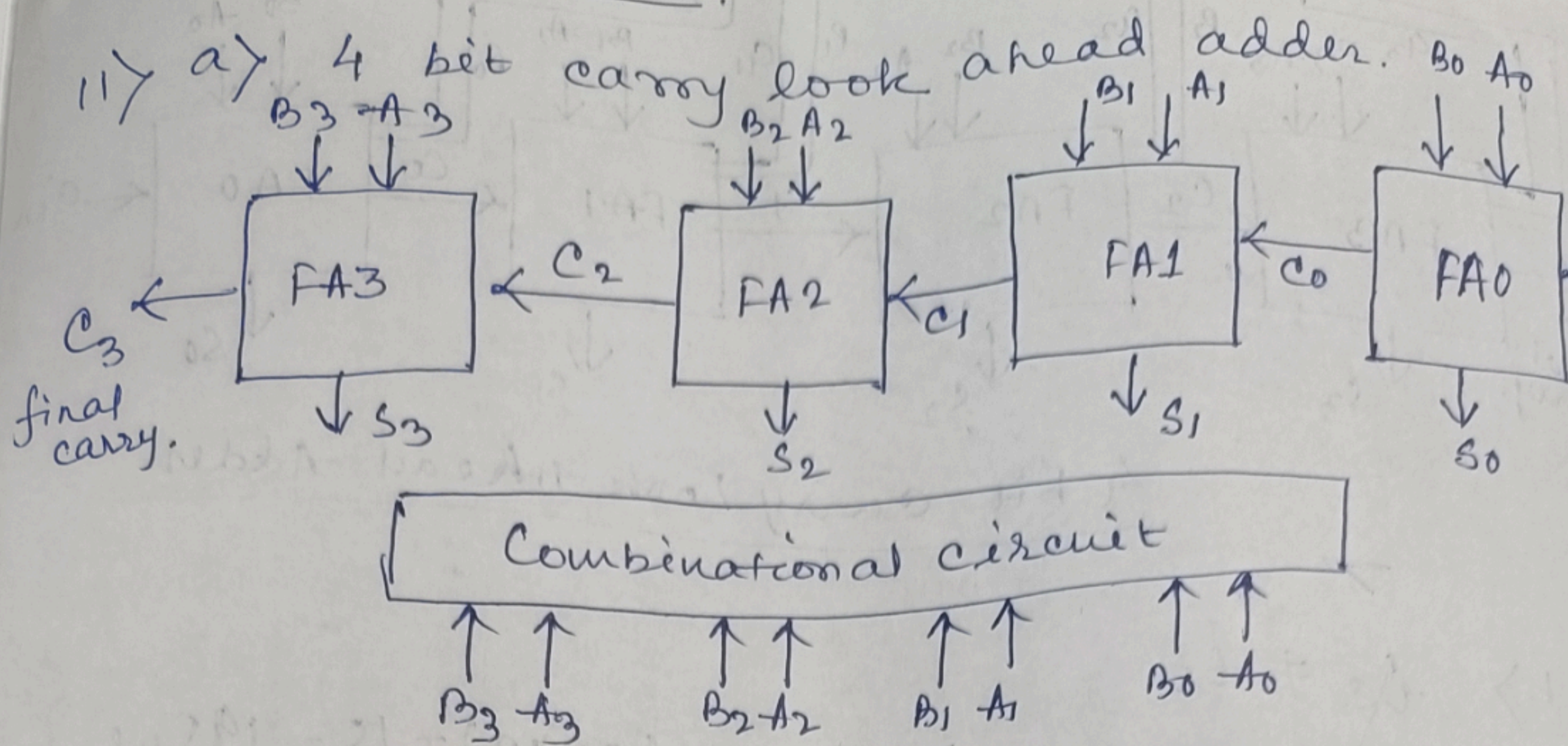
add m

$$\begin{array}{r} \boxed{1} \ 1 \ 1 \ 1 \quad 0 \ 0 \ 0 \text{ ---} \\ 0 \ 0 \ 1 \ 1 \\ \hline \boxed{0} \ 0 \ 1 \ 0 \quad 0 \ 0 \ 0 \ 1 \end{array}$$

Since the sign of A is 0 i.e. +ve,
hence no restoration reqd.

Ans: Accumulator (A) = 0 0 1 0 (remainder)
Dividend (Q) = 0 0 0 1 (quotient).

SECTION - C



now, ~~Ques 3~~

$$C_0 = \underbrace{A_0 B_0}_{G_0} + C_{in} \underbrace{(A_0 \oplus B_0)}_{P_0}$$

$$\Rightarrow \boxed{C_0 = G_0 + C_{in} P_0}$$

$$C_1 = A_1 B_1 + C_0 (A_1 \oplus B_1)$$

$$= G_1 + C_0 P_1$$

$$= G_1 + P_1 (G_0 + C_{in} P_0) \neq \text{G}$$

$$\boxed{C_1 = G_1 + P_1 G_0 + C_{in} P_0 P_1}$$

$$C_2 = A_2 B_2 + C_1 (A_2 \oplus B_2)$$

$$= G_2 + C_1 P_2$$

$$= G_2 + P_2 (G_1 + P_1 G_0 + C_{in} P_0 P_1)$$

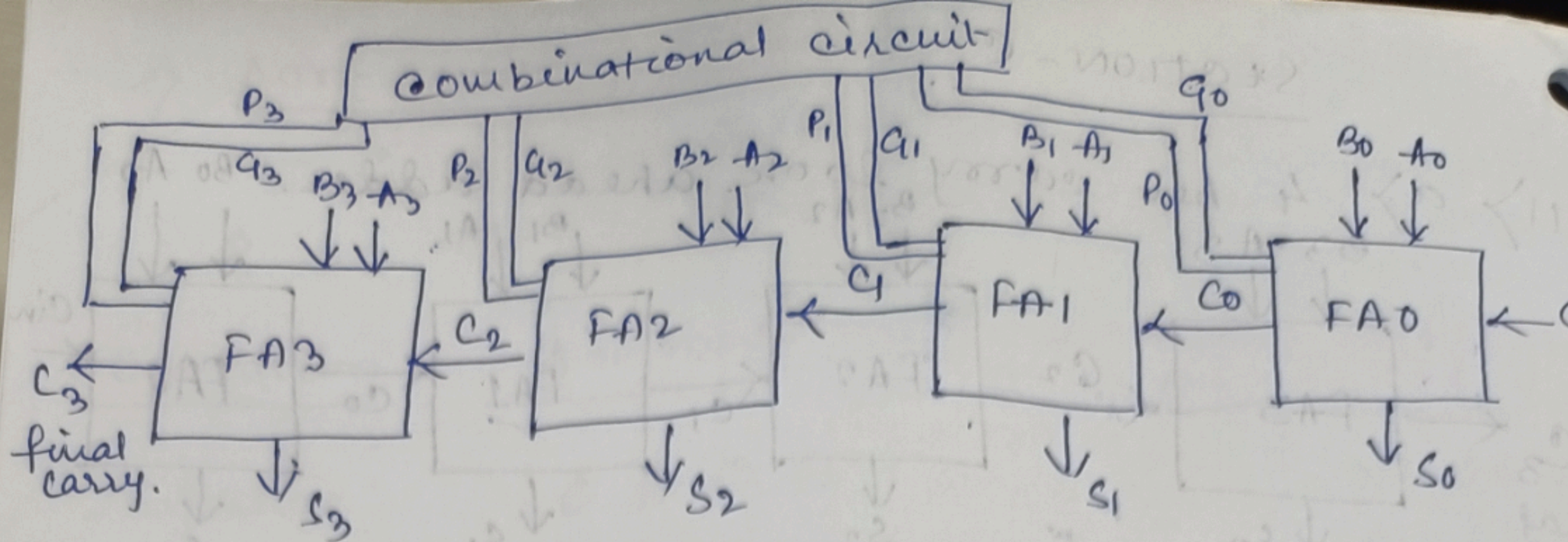
$$\boxed{C_2 = G_2 + G_1 P_2 + P_1 P_2 G_0 + C_{in} P_0 P_1 P_2}$$

$$C_3 = G_3 + C_2 P_3$$

$$= G_3 + P_3 (G_2 + G_1 P_2 + P_1 P_2 G_0 + C_{in} P_0 P_1 P_2)$$

$$C_3 = G_3 + G_2 P_3 + G_1 P_2 P_3 + P_1 P_2 P_3 G_0 + C_{in} P_0 P_1 P_2 P_3$$

The final carry i.e. C_3 is in terms of C_{in}



4 bit carry Look Ahead Adder

b) $t_c \neq t_s$

$$\therefore t_s = (n-1)t_c + t_s \Rightarrow (15 \times 12) + 15 = 195 \mu s$$

$$t_c = (n-1)t_c \Rightarrow 180 \mu s$$

$$\therefore \text{worst case delay} = 195 \mu s$$

Here, t_c is carry propagation delay

$$\text{given, } t_c = 12 \text{ ns}$$

$$\therefore t_c \text{ for 16 FAs} = (12 \times 16) \text{ ns} = 192 \mu s$$

t_s is sum propagation delay,

$$\text{given, } t_s \text{ for each adder} = 15 \text{ ns}$$

$$\therefore t_s \text{ for 16 FAs} = (15 \times 16) \text{ ns} = 240 \text{ ns}$$

$$\therefore \text{worst case delay} = (192 + 240) \text{ ns}$$

$$= 432 \text{ ns}$$

$$t_c \neq t_s$$

$$\therefore t_s = (n-1)t_c + t_s = 195 \mu s$$

$$t_c = (n-1)t_c = 180 \mu s$$

$$\therefore \text{worst case delay} = 375 \mu s$$